

2021 AIME II Problems/Problem 10

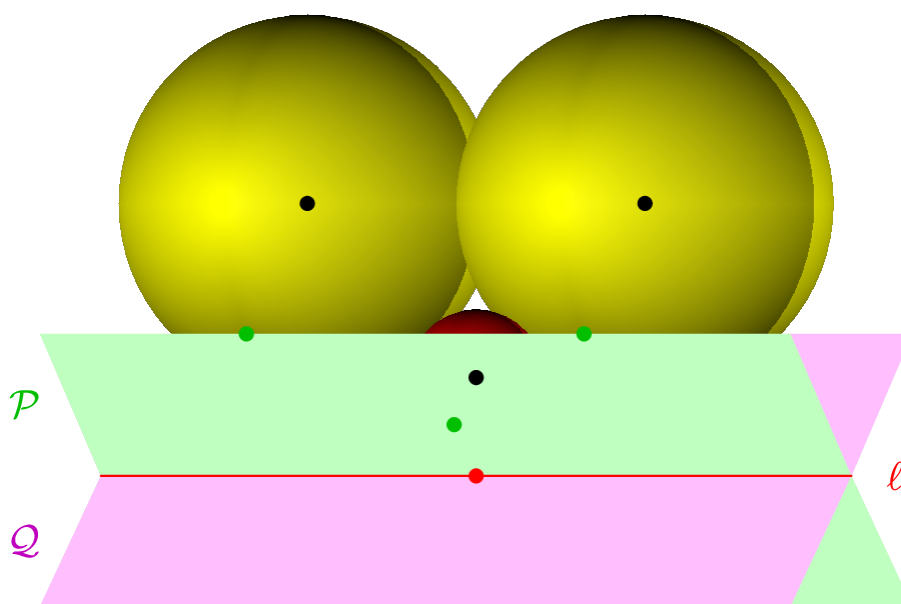
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Problem

Two spheres with radii 36 and one sphere with radius 13 are each externally tangent to the other two spheres and to two different planes $\mathcal{P}$ and $\mathcal{Q}$. The intersection of planes $\mathcal{P}$ and $\mathcal{Q}$ is the line ℓ . The distance from line ℓ to the point where the sphere with radius 13 is tangent to plane $\mathcal{P}$ is $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Diagram



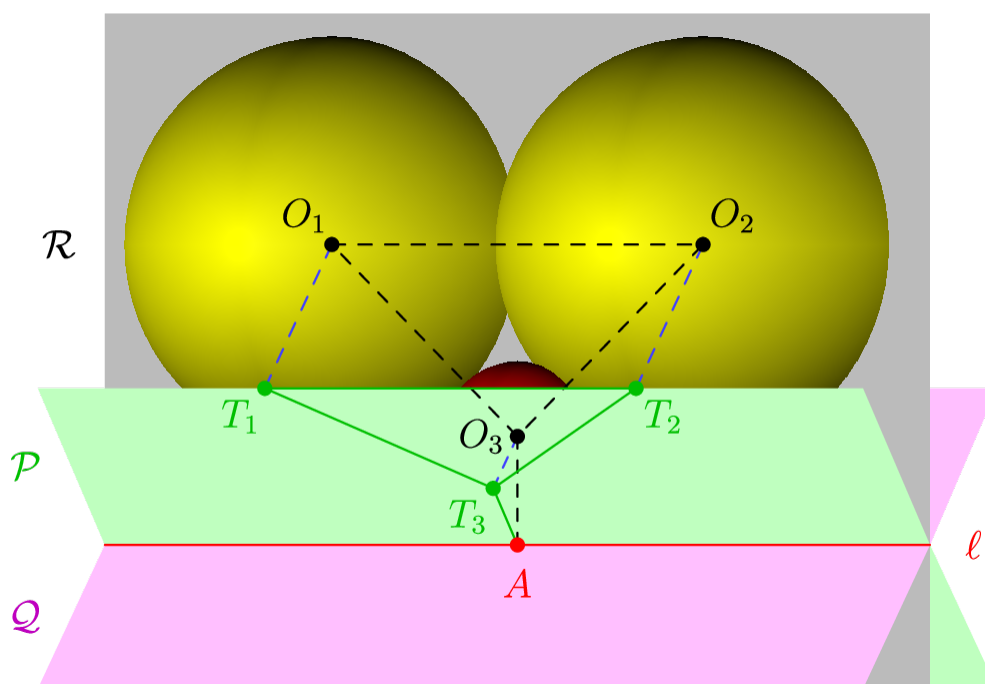
~MRENTHUSIASM

Solution 1 (Similar Triangles and Pythagorean Theorem)

This solution refers to the **Diagram** section.

As shown below, let O_1, O_2, O_3 be the centers of the spheres (where sphere O_3 has radius 13) and T_1, T_2, T_3 be their respective points of tangency to plane $\mathcal{P}$. Let $\mathcal{R}$ be the plane that is determined by O_1, O_2 , and O_3 . Suppose A is the

foot of the perpendicular from O_3 to line ℓ , so $\overleftrightarrow{O_3A}$ is the perpendicular bisector of $\overline{O_1O_2}$. We wish to find T_3A .



Note that:

1. In $\triangle O_1O_2O_3$, we get $O_1O_2 = 72$ and $O_1O_3 = O_2O_3 = 49$.
2. Both $\triangle O_1O_2O_3$ and $\overline{O_3A}$ lie in plane $\mathcal{R}$. Both $\triangle T_1T_2T_3$ and $\overline{T_3A}$ lie in plane $\mathcal{P}$.
3. By symmetry, since planes $\mathcal{P}$ and $\mathcal{Q}$ are reflections of each other about plane $\mathcal{R}$, the three planes are concurrent to line ℓ .
4. Since $\overline{O_1T_1} \perp \mathcal{P}$ and $\overline{O_3T_3} \perp \mathcal{P}$, it follows that $\overline{O_1T_1} \parallel \overline{O_3T_3}$, from which O_1, O_3, T_1 , and T_3 are coplanar.

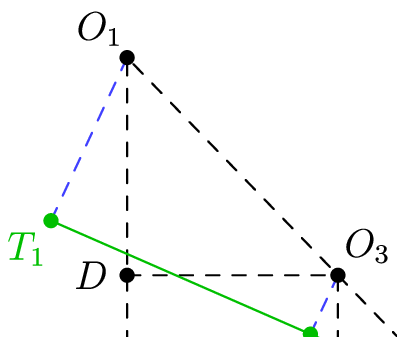
Now, we focus on cross-sections $O_1O_3T_3T_1$ and $\mathcal{R}$:

1. **In the three-dimensional space, the intersection of a line and a plane must be exactly one of the empty set, a point, or a line.**

Clearly, cross-section $O_1O_3T_3T_1$ intersects line ℓ at exactly one point. Furthermore, as the intersection of planes $\mathcal{R}$ and $\mathcal{P}$ is line ℓ , we conclude that $\overrightarrow{O_1O_3}$ and $\overrightarrow{T_1T_3}$ must intersect line ℓ at the same point. Let B be the point of concurrency of $\overrightarrow{O_1O_3}$, $\overrightarrow{T_1T_3}$, and line ℓ .

2. In cross-section $\mathcal{R}$, let C be the foot of the perpendicular from O_1 to line ℓ , and D be the foot of the perpendicular from O_3 to $\overline{O_1C}$.

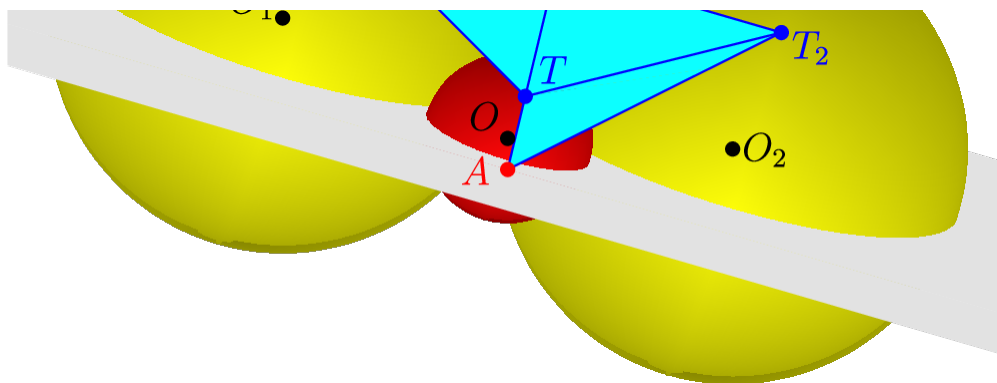
We have the following diagram:





~Ross Gao

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The isosceles triangle of centers O_1O_2O (O is the center of sphere of radii 13) has sides $O_1O = O_2O = 36 + 13 = 49$, and $O_1O_2 = 36 + 36 = 72$.

Let N be the midpoint O_1O_2 .

The isosceles triangle of points of tangency T_1T_2T has sides $T_1T = T_2T = 2\sqrt{13 \cdot 36} = 12\sqrt{13}$ and $T_1T_2 = 72$.

Let M be the midpoint T_1T_2 .

The height TM is $\sqrt{12^2 \cdot 13 - 36^2} = 12\sqrt{13 - 9} = 24$.

The tangents of the half-angle between the planes is $\frac{TO}{AT} = \frac{MN - TO}{TM}$, so $\frac{13}{AT} = \frac{36 - 13}{24}$,

$$AT = \frac{24 \cdot 13}{23} = \frac{312}{23} \implies 312 + 23 = \boxed{335}.$$

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Solution 4 (Coordinate Geometry)

Let the centers of the spheres be O_1, O_2, O_3 . We establish a 3D coordinate system with $\mathcal{P}$ being the $x - y$ plane, where O_1, O_2, O_3 have coordinates $(0, -36, 36), (0, 36, 36), (x_1, 0, 13)$, respectively. By the distance formula,

$$x_1^2 + 36^2 + 23^2 = 49^2 \implies x_1 = 24.$$

Let the equation of $\mathcal{Q}$ be $ax + by + cz + d = 0$. WLOG, we can assume $\sqrt{a^2 + b^2 + c^2} = 1$. Applying the distance formula on the 3 centers yields the system

$$\begin{aligned} |-36b + 36c + d| &= 36 \\ |36b + 36c + d| &= 36 \\ |24a + 13c + d| &= 13 \end{aligned}$$

By symmetry, $\mathcal{Q}$ is parallel to the y -axis, hence $b = 0$. Furthermore, since all spheres lie entirely on one side of $\mathcal{Q}$ we can choose the direction of the normal vector such that the inner expressions are positive. Dropping the absolute values gives

$$\begin{aligned} 36c + d &= 36 \\ 24a + 13c + d &= 13 \\ a^2 + c^2 &= 1 \end{aligned}$$

Solving yields $a = -\frac{1104}{1105}, c = -\frac{47}{1105}, d = \frac{41472}{1105}$. To find the intersection line ℓ between $\mathcal{Q}$ and $\mathcal{P}$, we set

$z = 0$, which gives the equation $x = -\frac{d}{a} = \frac{864}{23}$. The small sphere touches $\mathcal{P}$ at $(24, 0, 0)$. The distance from this point of tangency to ℓ is thus

$$\frac{864}{23} - 24 = \frac{312}{23}$$

The requested sum is $312 + 23 = \boxed{335}$.

~Pieandpi31415926535897932

Video Solution by Interstigation

<https://youtu.be/bQ3KdG4xH0A>

~Interstigation

See Also

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